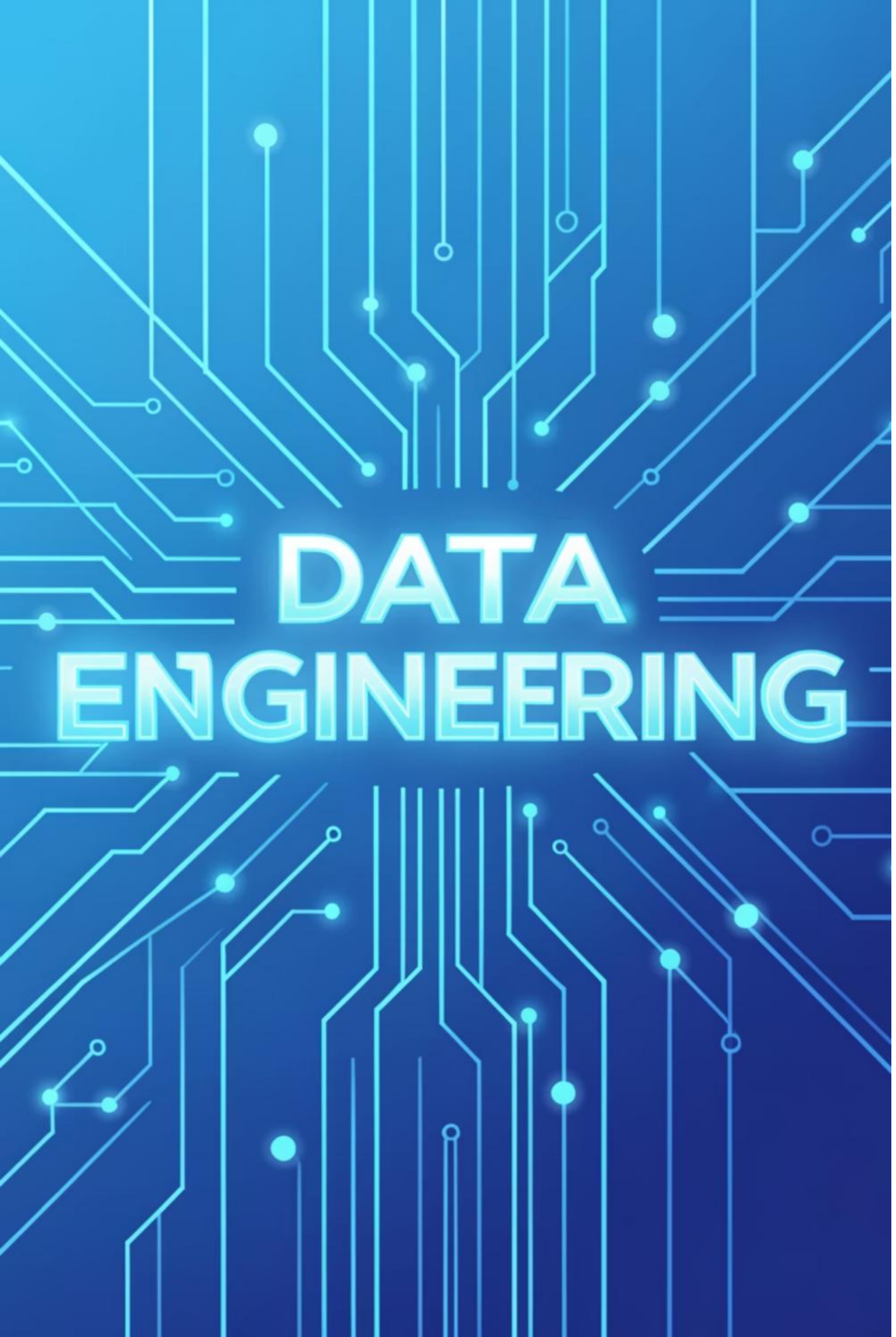

Data Engineering

Spring '26 – DS463

Feb 23, 2026

Lecture 16

Logistics: Your Course Essentials Essentials

Class Timings

Monday (LH6), Tuesday (LH8): 11:30 AM AM – 12:20 PM

Friday (LH8): 11:00 AM – 11:50 AM

Credit Hours

2 hours (Lectures) + 1 hour (Practical (Practical Session)

Office Hours

Monday, Tuesday: 10:00 AM – 11:00 AM, AM, S18 2nd Floor NAB or by appointment.

Contact & Support

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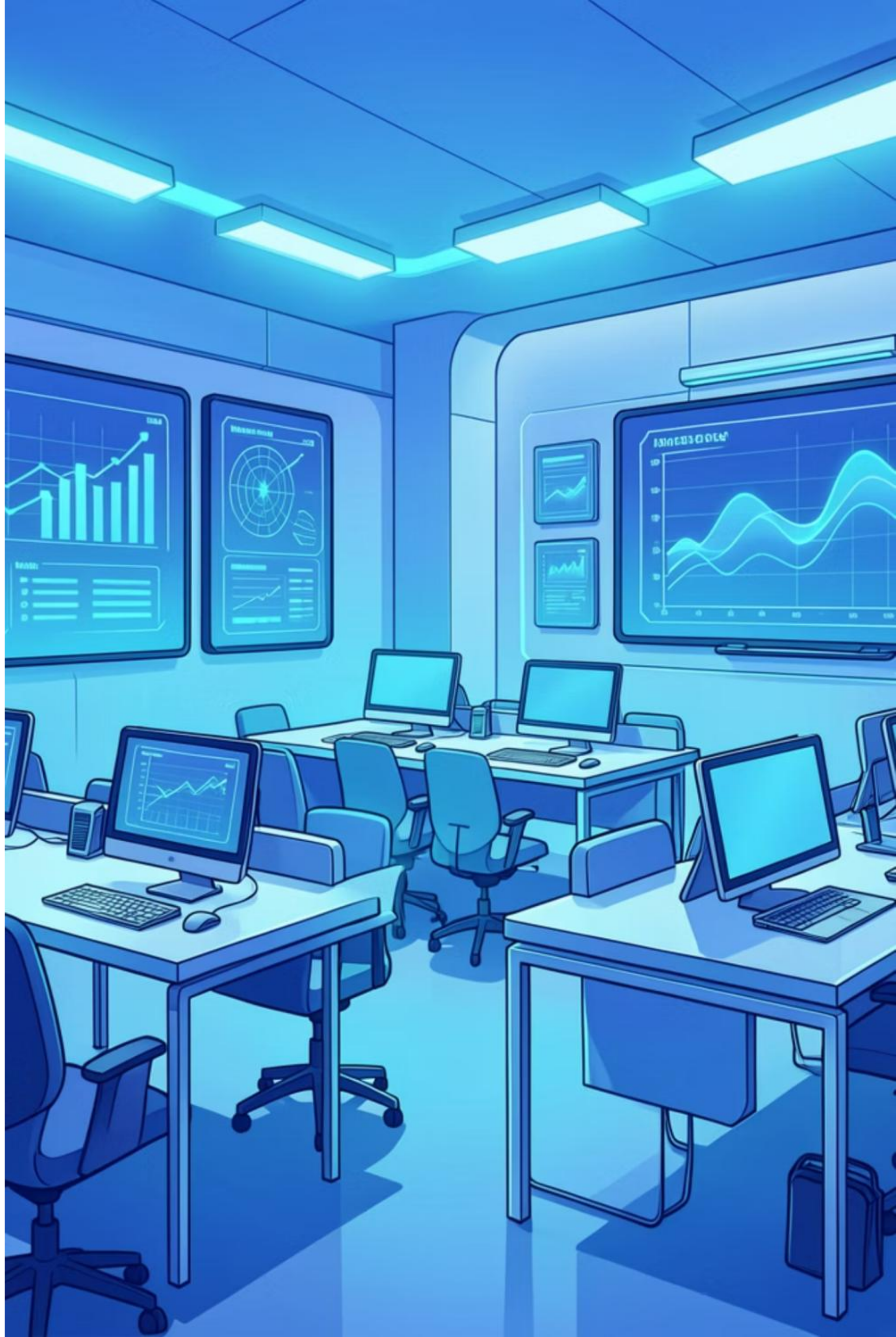
Email: farah.saeed@giki.edu.pk

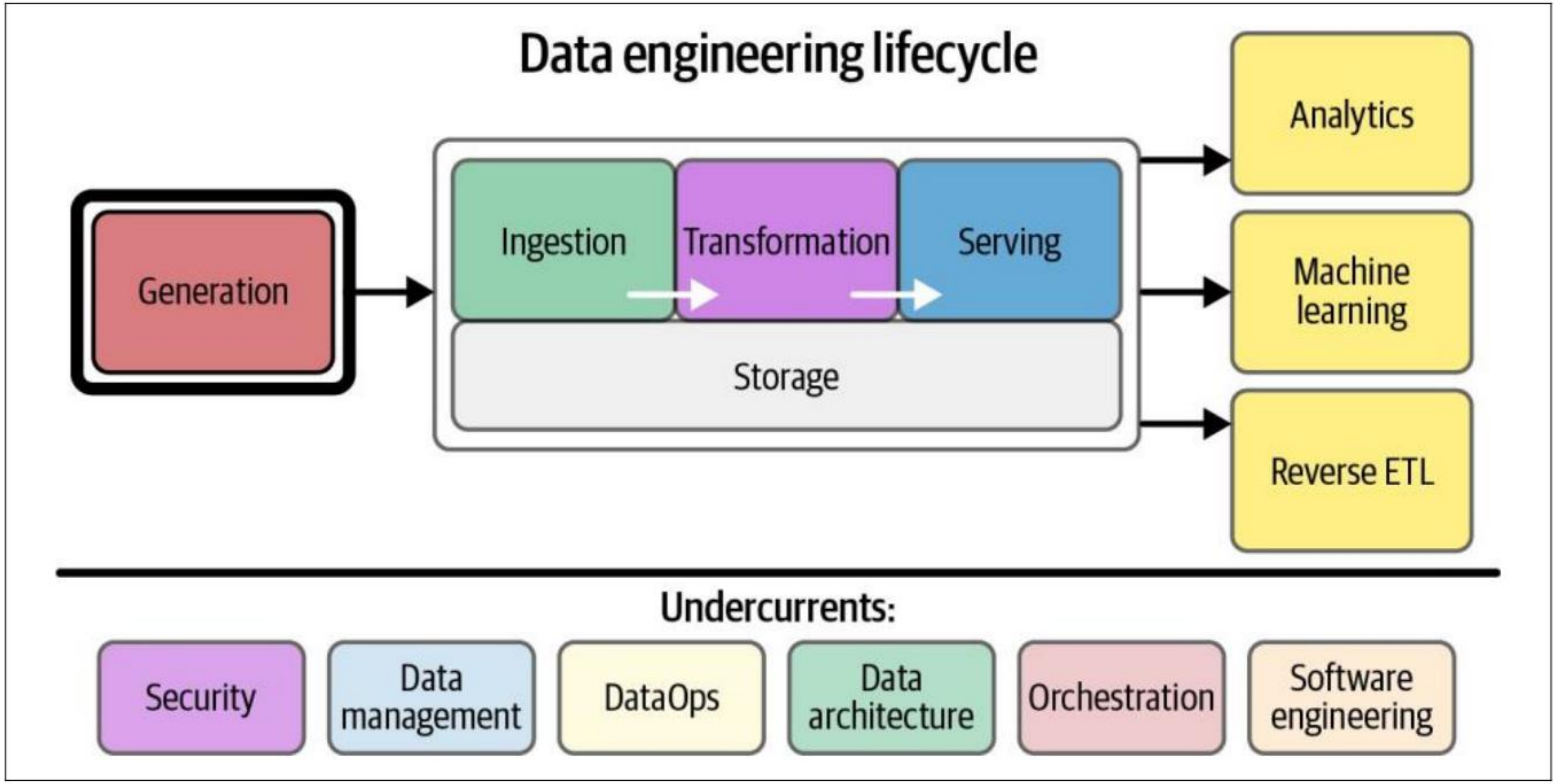
WhatsApp: Reach out for quick queries



Major Architecture Concepts

- **Domain and Services**
- **Distributed Systems, Scalability and Design for failure**
- **Tight vs Loose Coupling**
- **User Access**
- **Event Driven Architecture**
- **Brownfield vs Greenfield Projects**

Generation in Source Systems



Sources of Data

What is Data?

Data is a collection of raw facts and figures.

It is unorganized and has no meaning by itself.

It becomes meaningful only when given context.

Sources of Data

How is Data created?

Analog Data

Created in the physical world.

Examples:

- Speaking in a conversation
- Sign language
- Writing on paper
- Playing a musical instrument

Often temporary (transient).

Example: A verbal conversation that is not recorded and later forgotten.

Digital Data

a) Converting Analog to Digital

Real-world data is converted into digital form.

Example: A mobile app converting speech into text messages.

b) Born-Digital Data

Created directly by digital systems.

Examples:

Online credit card transactions, Ecommerce orders, Database records.

Source systems: Main Ideas

Files and Unstructured Data

What is a File?

A file is a sequence of bytes stored on a disk.

Applications use files to store data.

Files can contain:

- Settings or parameters
- Logs and events
- Images
- Audio

Files are a common way to share data between systems.

Files can contain different levels of structure:

Structured data

Excel

CSV

Semistructured data

JSON

XML

Sometimes CSV

Unstructured data

TXT

Sometimes CSV Others (e.g. Parquet, ORC)

Source systems: Main Ideas

Application Programming Interface APIs

What is an API?

A standard way for systems to exchange data.

Allows programs to communicate directly. APIs make data ingestion easier.

Data can be accessed programmatically.

In Practice

APIs can still be complex.

Engineers often need to:

Handle authentication

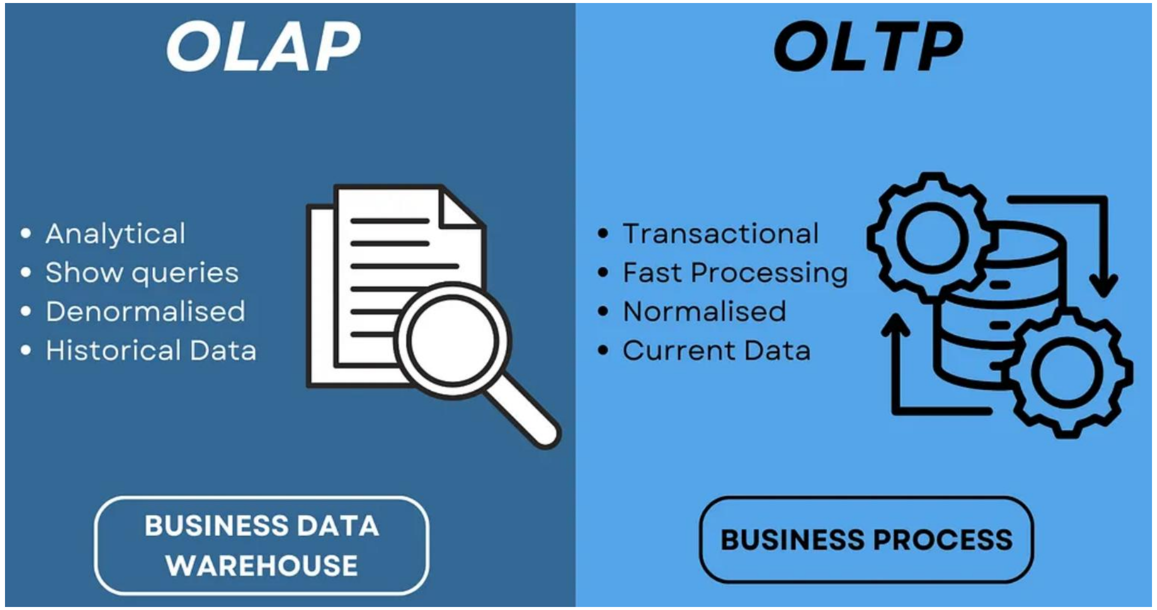
Manage rate limits

Source systems: Main Ideas

Data warehouse

Characteristics

- Separates OLAP from OLTP
- OLTP = transactional systems (production databases).
- OLAP = analytics systems
- Separating them:
  - Protects production systems.
  - Improves analytics performance.



https://blog.dataengineerthings.org/transactional-vs-analytical-a-deep-dive-into-oltp-and-olap-for-data-professionals-feb013a7264f

Source systems: Main Ideas

Change Data Capture

CDC is a method for tracking changes in a database.

It captures every:

- Insert (new record added)
- Update (existing record modified)
- Delete (record removed)

Instead of copying the whole database, it captures only the changes.